

# PROJECT DATA SHEET

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**JOB/PROJECT :** MULTI FAITH TEMPORARY WORKS**JOB/PROJECT NO :** C8676      **SHEET NO :** COVER**GENERAL :****CLIENT :** WILLMOTT DIXON CONSTRUCTION LTD**ARCHITECT :** N/A**AUTHORITY :** EXETER CITY COUNCIL**JOB/PROJECT DESCRIPTION :** SCAFFOLDING MAT DESIGN**PROPOSED USE:** SCAFFOLDING MAT**OVERALL SIZE :** AS CALCS**No. OF STOREYS :** N/A**TYPE OF SOIL INVESTIGATION :** N/A**BEARING STRATA :** N/A**TYPE OF FOUNDATION :** N/A**MAXIMUM SAFE BEARING PRESSURE**75 kN/m<sup>2</sup>**STRUCTURAL MATERIALS USED :****STANDARDS AND CODES USED IN DESIGN :**

|                                                               |                  |                          |           |                               |                                     |
|---------------------------------------------------------------|------------------|--------------------------|-----------|-------------------------------|-------------------------------------|
| General actions:<br>Densities, self weight &<br>imposed loads | BS EN 1991-1-1   | <input type="checkbox"/> | Steelwork | BS EN 1993 (EC3)              | <input type="checkbox"/>            |
|                                                               |                  |                          | Timber    | BS EN 1995 (EC5)              | <input type="checkbox"/>            |
| General actions:<br>Actions on structures<br>exposed to fire  | BS EN 1991-1-2   | <input type="checkbox"/> | Masonry   | BS EN 1996 (EC6)              | <input type="checkbox"/>            |
| Snow Loads                                                    | BS EN 1991-1-3   | <input type="checkbox"/> | Others    | TG 20:21 OPERATIONAL<br>GUIDE | <input checked="" type="checkbox"/> |
| Wind Loads                                                    | BS EN 1991-1-4   | <input type="checkbox"/> |           |                               | <input type="checkbox"/>            |
| Concrete                                                      | BS EN 1992 (EC2) | <input type="checkbox"/> |           |                               | <input type="checkbox"/>            |

**AUTHORISATION OF CALCULATION SHEETS AND BENDING SCHEDULES :**

| CALCULATION SHEET NOS.       | CHECKED BY | DATE      |
|------------------------------|------------|-----------|
| Sheet nos: S01—S02 inclusive | A.C. HEARD | 30/01/24. |
|                              |            |           |
|                              |            |           |



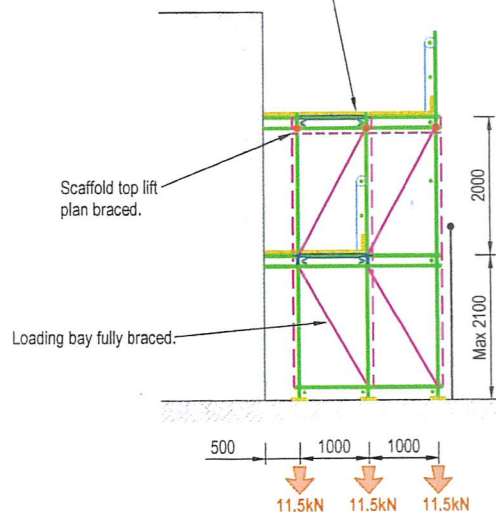
## Calculations

|          |                                       |             |       |           |          |
|----------|---------------------------------------|-------------|-------|-----------|----------|
| Contract | MULTI FAITH CENTRE<br>TEMPORARY WORKS | Job No.     | C8676 | Sheet No. | 501      |
|          |                                       | Designed by | D.P.H | Date:     | JAN 2024 |

### DESIGN OF SCAFFOLDING MAT

From information provided by client OPTIMA scaffold designs LLP drawing NO. 23/OPT/18718-001 (sheet 2) the maximum leg load = 11.5kN.

Loading bay board bearers at max 300mm c/c, secured using clips.



**TYPICAL SECTION**  
(LOADING BAY SCAFFOLD)

From TG 20:21 Operational guide for leg loads less than 20kN the minimum sole board size is 450x225x38mm. These are to be used in tandem with 150x150x10mm base plates as shown in the figure below.

| Leg load                 | Minimum sole boards                                            | Figure             |
|--------------------------|----------------------------------------------------------------|--------------------|
| Maximum leg load<br>< 20 | 450x225x38mm                                                   | 5.1                |
| Maximum leg load<br>20+  | Double boards<br>450x225x38mm                                  | 5.6                |
| Combined leg load<br>20+ | 750x225x75mm, or<br>Two pairs of double<br>450x225x38mm boards | 5.2 (a)<br>5.2 (b) |

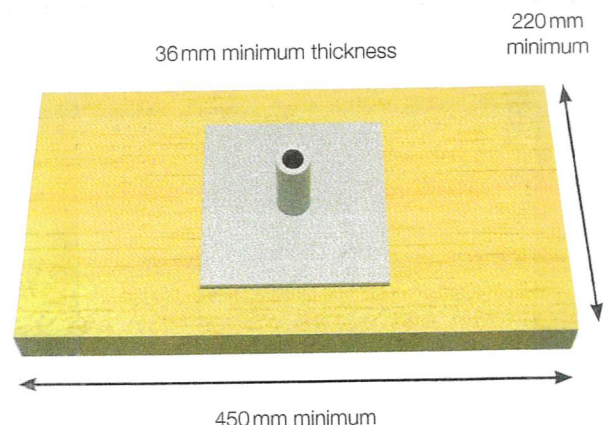


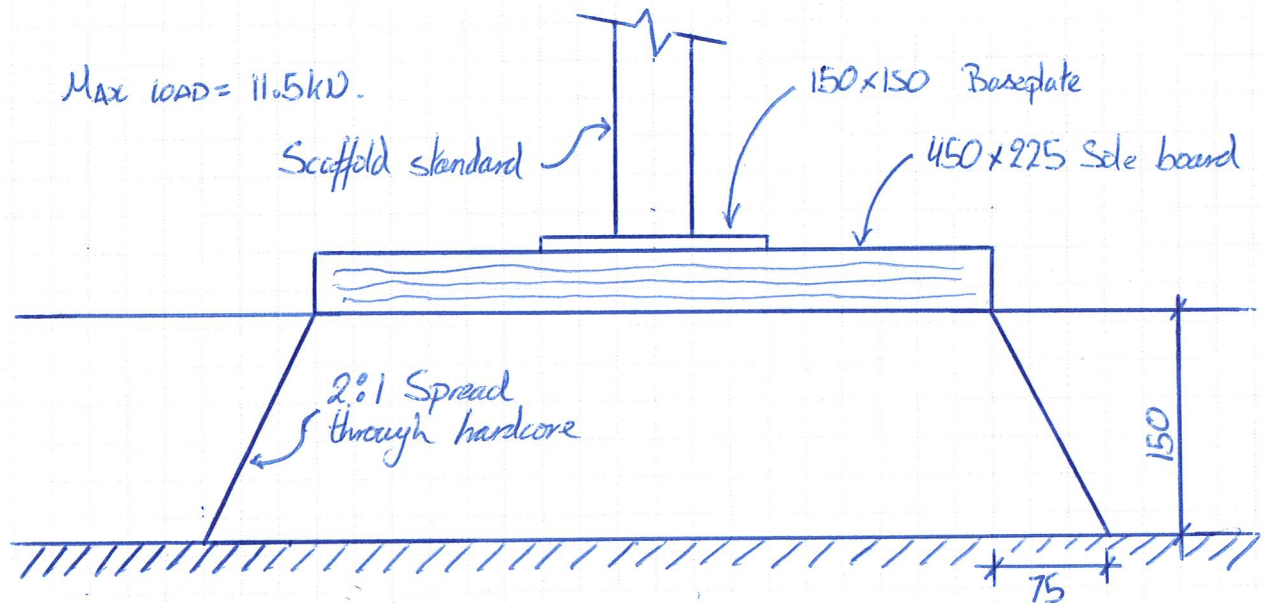
Table 5.1: minimum sole boards for TG20 compliant scaffolds.





## Calculations

|          |                                       |             |       |           |          |
|----------|---------------------------------------|-------------|-------|-----------|----------|
| Contract | MULTI FAITH CENTRE<br>TEMPORARY WORKS | Job No.     | C8676 | Sheet No. | S02      |
|          |                                       | Designed by | D.P.H | Date:     | JAN 2024 |



Max load = 11.5kN.

Scaffold standard

150x150 Baseplate

450x225 Sole board

2:1 Spread  
through hardcore

150

75

$$\text{Bearing pressure under Sole board on mat} = \frac{11.5}{0.45 \times 0.225} = 114 \text{ kN/m}^2$$

$$\text{Permissible Bearing pressure on mat} = 200 \text{ kN/m}^2 > 114 \text{ kN/m}^2 \therefore \text{Ok.}$$

$$\text{Max permissible Bearing pressure under mat on ground} = 75 \text{ kN/m}^2$$

$$\therefore \text{Depth of mat required} - \frac{11.5}{(0.45+x) \times (0.225+x)} = 75 \quad x = 0.07 \text{m} = 70 \text{mm}$$

$$\therefore \text{Minimum mat depth to achieve } 75 \text{ kN/m}^2 = 70 \text{mm}$$

$\therefore$  USE Nominal 150dp mat

$\therefore$  USE 150x150 x  $\pm 10$ mm BASEPLATE, ON 450x225 x 38mm  
SOLE BOARD, ON 150mm THICK HARDCORE MAT